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RESEARCH**

APPLICATION NUMBER:

209478Orig1s000

**CLINICAL PHARMACOLOGY AND
BIOPHARMACEUTICS REVIEW(S)**

Clinical Pharmacology Review

Drug Substance:	Spironolactone
NDA Number:	209,478
Proposed Brand Name:	CaroSpir®
Dosage Form:	Oral Suspension
Dosage Strength:	25 mg/5 mL
Indication:	(b) (4) heart failure, (b) (4) hypertension
Receipt Date:	10/04/2016
Applicant:	CMP PHARMA INC.
Primary Reviewer:	Xiaolei Pan, Ph.D.
Secondary Reviewer:	Martina Sahre, Ph.D.
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OCP Division:	DCP 1
OND Division:	DCRP
Submission Type:	505(b)(2)

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1 Executive Summary

The applicant submitted a 505(b)(2) for spironolactone oral suspension, 25 mg/5 mL. The submission is relying on the finding of safety and efficacy of Aldactone® (NDA 012,151), which is FDA-approved for the treatment of primary hyperaldosteronism, edematous conditions, heart failure, hypokalemia, nephrotic syndrome and essential hypertension.

The applicant is relying on two relative bioavailability studies, one conducted with the 100 mg tablets as reference (Study 064-15) and one with 25 mg tablets as reference, to provide the comparison between spironolactone oral suspension and the listed drug (LD, Aldactone®, NDA 12,151). A food-effect study was also conducted.

The applicant seeks approval of the investigational drug for (b) (4)

The results of the studies show that the bioavailability of spironolactone suspension is 15% and 37% higher than the LD at 25 mg and 100 mg tablet strengths, respectively. A product-specific OGD guidance on spironolactone exists and bioequivalence is established for the highest Aldactone® dose strength of 100 mg, while biowaivers based on *in vitro* dissolution and proportional similarity are sought for lower dose strengths. One assumption for issuing of waivers for lower dose strengths is dose proportionality across the dose range. If that assumption were true, then the increase in exposure would have been expected to be approximately similar between the two studies. The applicant asserted that the assumption of proportional similarity, i.e. that the increase in amount of excipients between the two tablet strengths is similar to the increase in spironolactone strength, cannot hold, due to the tablet sizes being the same. Given the lack of information on proportional similarity of dose strengths and dose proportionality the published literature, we cannot exclude the possibility that increases in exposure at doses above 100 mg might be larger than what was observed at the 100 mg comparison. Therefore, the applicant will need to submit information that characterizes dose proportionality across the clinically relevant dose range (25 to (b) (4) mg), as well as assess relative bioavailability for all dose strengths. In the absence of that information, (b) (4) spironolactone suspension should be limited to 25 to 100 mg. Additionally, to overcome the increased exposure, a 25% dose reduction would be needed for both doses to adjust for the higher exposure to spironolactone from suspension between 25 and 100 mg. The applicant proposed to address this topic by (b) (4)

However, this approach would still not address the concern about dose proportionality.

1.1 Recommendations

The Office of Clinical Pharmacology/Division of Clinical Pharmacology 1(OCP/DCP1) has reviewed the original NDA 209,478. The data are acceptable to support approval provided that a mutual agreement for the revision of label language can be reached between the applicant and the division. Key recommendations are as following:

- The approved indications are limited to heart failure, hypertension, and edema caused by (b) (4) cirrhosis (b) (4) within a dose range of 18.75 mg to 75 mg.
- A 25% dose reduction for spironolactone oral suspension relative to the LD.

1.2 Post Marketing Requirement

None

1.3 Summary of Clinical Pharmacology and Biopharmaceutics Findings

The development program for spironolactone oral suspension was designed to bridge to the listed drug Aldactone® 100 mg tablets (NDA 12,151) for efficacy and safety, by conducting a pivotal relative bioavailability study, comparing 100 mg tablets to 100 mg suspension, as well as a pilot relative BA study, comparing 25 mg tablet to suspension, and a food effect study. The relative bioavailability studies showed that spironolactone suspension is not bioequivalent to spironolactone tablets. Moreover, when comparing the results from the pivotal and the pilot relative BA studies (064-15 and 063-15, respectively), the resulting increase in exposure was higher than expected given the results from the pilot assessment (25 mg spironolactone tablets vs suspension).

The key clinical pharmacology findings were as follows:

Study 063-15: The relative bioavailability of the spironolactone suspension is 115.2% (95% CI: 104.2% -127.4%) as compared to spironolactone 25 mg tablet; while the ratio of C_{max} is approximately 121.7% (95% CI: 101.4% - 146.1%) between 25 mg spironolactone suspension and tablet.

Study 064-15: The relative bioavailability of spironolactone suspension is approximately 137.0% (95% CI: 122. 5% -153.3%) as compared to spironolactone 100 mg tablet; while the ratio of C_{max} is approximately 165.6% (95% CI: 150.4% - 182.3%) between 100 mg spironolactone suspension and tablet.

Study 084-15: Food increased the exposure of spironolactone by approximately 90.5 % (95%CI: 73.9% -108.6%) for the suspension; while it decreased C_{max} by 21.6% (95% CI: 30.4%-11.6%).

2 Question-Based Review

This is an abridged version of the question-based review. For detailed information about spironolactone, refer to the original NDA 12,151.

2.1 General attributes of the drug and drug product

Spironolactone oral suspension is a white to off-white opaque suspension provided in a dose strength of 25 mg in 5 mL. It is available in a 4 ounce (118 mL) bottle or a 16 ounce (473 mL) bottle.

2.1.1 What are the highlights of the chemistry and physical-chemical properties of the drug substance and the formulation of the drug product and listed drug (LD) as they relate to clinical pharmacology and biopharmaceutics review?

The inactive ingredients in spironolactone oral suspension include xanthan gum (b) (4), simethicone emulsion (b) (4), citric acid anhydrous and sodium citrate dehydrate (b) (4), sorbic acid and potassium sorbate (b) (4), sodium saccharin, Magnasweet 100, glycerin and banana flavor (b) (4) and purified water.

2.1.2 Are additional instructions for preparation and dispensation of the final solution required to ensure accurate dosing?

The proposed drug label of the spironolactone oral suspension directs to shake well before use.

In addition, the study protocols recorded that the administration vessel was rinsed several times in order to prevent loss of medication. Instructions for suspension administration were as following:

- Test product will be administered with a graduated syringe.
- 5 mL of suspension will be loaded in the syringes well in advance prior to dosing on the day of dosing.
- The suspension in the syringe will be directly administered to the subjects.
- The syringe will be rinsed with water (approximately 10 mL from 240 mL of dosing water) and will be administered to the subjects. This procedure will be followed till the syringe is free of the contents of the medication. Then the remaining quantity of water (of the 240 mL) will also be administered to the subject.

Given that the study instructions were to rinse the syringe with water, this information should be added to the label.

2.1.3 What are the proposed mechanism(s) of action and therapeutic indication(s)?

Spironolactone and its active metabolites (e.g., canrenone and 7- α -(thiomethyl) spironolactone (TMS)) are pharmacologic antagonists of aldosterone, acting primarily through competitive binding of receptors at the aldosterone-dependent sodium-potassium exchange site in the distal convoluted renal tubule. Through its action in antagonizing the effect of aldosterone, spironolactone causes increased amounts of sodium and water to be excreted, while potassium is retained.

Spironolactone is indicated for (b) (4) management of edema associated with (b) (4) liver cirrhosis, (b) (4) hypertension; (b) (4) and severe heart failure (NYHA class III and IV).

2.1.4 What are the proposed dosage(s) and route(s) of administration?

(b) (4). Spironolactone oral suspension is proposed to be given via oral administration.

Table 1. Proposed dosing for Spironolactone oral suspension

Indication	Dose
(b) (4)	(b) (4)
Edema	(b) (4)
Essential hypertension	
(b) (4)	
Severe heart failure (b) (4)	(b) (4)

The reviewer recommends a dose reduction for spironolactone oral suspension relative to the LD and limiting the dosage regimen between 25 mg and 100 mg, Please refer to Sections 1.1 and 2.3 for details of recommendations and rationale.

2.2 General clinical pharmacology

2.2.1 What are the design features of the clinical pharmacology and clinical studies used to support dosing or claims?

This NDA contains three clinical pharmacology studies. They are summarized in Table 2.

Table 2. Summary of clinical pharmacology studies

Study	Reference	N	Fed/Fasted
063-15 (Pilot)	1 x 25 mg Aldactone tablets	14	Fasted
064-15 (Pivotal)	100 mg Aldactone tablets	56	Fasted
084-15 (Food-Effect)	100 mg Aldactone tablets	23	Fed

2.2.2 Are the active moieties in the plasma appropriately identified and measured to assess pharmacokinetic parameters?

The moieties that were analyzed were spironolactone, canrenone and in study 063-15 also TMS. All methods were validated and performed within acceptable limits during study sample analysis. Refer to section 2.4 for details about the analytical methods.

2.3 General Biopharmaceutics

2.3.1 What is the relative bioavailability of the proposed oral solution relative to the listed drug?

Two relative bioavailability studies were conducted between the spironolactone oral suspension and the LD in study 063-15 and 064-15 (see **Error! Reference source not found.**). The study design is described in the individual study reviews. Briefly, study 063-15 compared relative bioavailability of 25 mg spironolactone suspension with 25 mg tablet in healthy subjects under fasting condition; while study 064-15 compared relative bioavailability study of 100 mg spironolactone suspension with 100 mg tablets.

Spironolactone oral suspension is not bioequivalent to the LD at both 25 mg and 100 mg, because the 90% CIs of the geometric mean ratios of AUC did not fall within the 80 and 125% bioequivalence limits. Bioavailability of spironolactone suspension is approximately 15% higher compared to spironolactone 25 mg tablets and 37% higher compared to spironolactone 100-mg tablets. C_{max} increases 22 and 66%, respectively (Table 3).

Table 3. Ratio and 90% confidence intervals of test versus reference for spironolactone in studies 063-15 and 064-15.

Parameters (Unit)	Study 063-15			Study 064-15		
	(T/R)(%)	CV	90% CI	(T/R)(%)	CV	90% CI
C _{max} (ng/mL)	121.7%	26.3%	101.4% - 146.1%	165.6%	31.1%	150.4% - 182.3%
AUC _{0-t} (ng*hr/mL)	114.0%	14.4%	103.0% -126.1%	137.4%	33.7%	123.8% - 152.4%
AUC _{0--inf} (ng*hr/mL)	115.2%	14.3%	104.2% - 127.4%	137.0%	36.6%	122.5% - 153.3%

2.3.2 Are the observations from both relative bioavailability studies expected?

C_{max} and AUC increased less than dose-proportionally from 25 mg to 100 mg for the LD, spironolactone tablets, when comparing studies 063-15 and 064-15 (Table 4).

Table 4. Summary of pharmacokinetic parameters for spironolactone in study 063-15 and study 064-15.

PK Parameters	Study 063-15		Study 064-15	
	25 mg (Test)	25 mg (LD)	100 mg (Test)	100 mg (LD)
C_{max} (ng/mL)	47.4 (40.7%)	40.4 (35.6%)	127.0 (41.6%)	76.7 (53.6%)
AUC_{0-t} (ng*hr/mL)	76.7 (35.1%)	68.8 (33.6%)	290.1 (47.5%)	211.2 (56.9%)
AUC_{0-inf} (ng*hr/mL)	81.6 (35.0%)	72.6 (33.4%)	299.5 (50.3%)	218.6 (57.6%)

One assumption when studying only the highest labeled tablet strength and waiving lower strengths based on in vitro dissolution testing, is that there is a proportional increase in PK with increasing dose. If the pharmacokinetics of spironolactone following administration of Aldactone® tablets were truly dose-proportional, the expectation would have been that the relative increase in bioavailability for oral suspension compared to 25 and 100 mg tablets would have been similar, which was not the case here. The reason for this observation is not entirely clear. It could be due to chance alone, or due to other factors in study conduct or sample analysis. An inspection by OSIS, was requested for both studies, to assess whether study conduct could have had an impact. Additionally, the label for the LD does not contain any information on the dose proportionality, and the data reported in the published literature does not appear sufficient to assume dose proportionality for spironolactone. Thus, an information request (IR) was sent to the applicant on Dec 14, 2016, to request justification for the observed lack of dose proportionality.

The applicant's response (Jan 13, 2017) suggested that Aldactone® 25 mg and 100 mg tablets are not compositionally proportional because they are both the same physical size, which may partially explain the difference in the relative bioavailability results between the two studies. In the absence of right to reference to the formulations of Aldactone®, it is not possible to corroborate that assertion.

The lack of information about dose proportionality and the observed differences regarding dose proportionality raise a concern that there may be a potential for greater difference in exposure between the suspension and tablets at doses higher than 100 mg.

(b) (4) a labelled 100% increase in exposure due to food was not accompanied by a dose adjustment (b) (4)

Nevertheless, we are concerned about the potential combined effect of an increase in exposure due to food and the higher bioavailability with the suspension.

In light of these issues, the applicant should submit information about the dose-proportionality in the clinically relevant dose range of 25 to (b) (4) mg, and the relative bioavailability studies of the oral suspension compared to the tablets at the respective dose strengths should be conducted.

In the absence of the aforementioned information, the approval of spironolactone suspension should be limited to between 25 mg and 100 mg (b) (4)

The highest labeled dose for edema is (b) (4) mg. (b) (4)

(Response to IR letter, received May 30th, 2017). However, given the lack of information about dose proportionality, explained above, this approach is not considered feasible by the Division.

2.3.3 Is a dose adjustment required to account for the higher exposure from spironolactone oral suspension?

The observed higher relative bioavailability of spironolactone oral suspension compared to the LD at both 25 mg and 100 mg demonstrated a need for dose reduction when transitioning patients from stable therapy with the tablet formulation to your product. However, as mentioned in section 2.3.1, the results indicate that the relative bioavailability of spironolactone is approximately 15% higher compared to 25 mg spironolactone tablets and 37% higher compared to 100 mg spironolactone tablets. A 25% dose reduction relative to the LD would allow AUC for both dose strengths to fall approximately in the 80-125% range compared to tablets. The estimated geometric mean ratios for the C_{max} , AUC_{0-t} and AUC_{0-inf} between spironolactone oral suspension and the LD after dose adjustment are summarized in Tables Table 5 and Table 6 .

The results show that the 18.75 mg spironolactone oral suspension is estimated to be bioequivalent to the 25 mg LD; while the 75 mg spironolactone oral suspension is estimated to be bioequivalent to the 100 mg LD. Thus, a 25% dose reduction of spironolactone oral suspension relative to the LD could be able to successfully alleviate the relative higher exposure of spironolactone from spironolactone suspension within this dose range. In addition, the dose of 18.75 mg (3.75 mL) is further rounded up to 20 mg (4mL) for the consideration of applicability during clinical practice.

Table 5. The estimated ratio and 90% confidence intervals for spironolactone after a 25% dose reduction for 25 mg spironolactone oral suspension relative to 25 mg LD.

Parameters	Geometric Least Square Means			Estimated Ratio after 25% Dose Reduction	90% CI
	Spironolactone Suspension (25 mg)	Spironolactone Suspension (18.75 mg)	Aldactone® (LD 25 mg)		
C_{max} (ng/mL)	47.4	37.9	40.4	96.1%	80.3% – 115.2%
AUC_{0-t} (ng*hr/mL)	76.7	61.4	68.8	90.7%	81.9% - 100.3%
AUC_{0-inf} (ng·hr/mL)	81.6	65.3	72.6	91.6%	82.9% - 101.3%

Table 6. The estimated ratio and 90% confidence intervals for spironolactone after a 25% dose reduction for 100 mg spironolactone oral suspension relative to 100 mg LD.

Parameters	Geometric Least Square Means			Estimated Ratio after 25% Dose Reduction	90% CI
	Spironolactone Suspension (100 mg)	Spironolactone Suspension (75 mg)	Aldactone® (LD 100 mg)		
C _{max} (ng/mL)	127.0	101.6	76.7	132.5%	120.3% - 145.8%
AUC _{0-t} (ng*hr/mL)	290.1	232.1	211.2	109.9%	99.1% - 121.9%
AUC _{0-inf} (ng*hr/mL)	299.5	239.6	218.6	109.6%	98.0% - 122.6%

2.3.4 What is the effect of food on the bioavailability (BA) of the drug from the dosage form? What dosing recommendation should be made, if any, regarding administration of the product in relation to meals or meal types?

Food effect on the bioavailability of spironolactone was assessed in study 084-15 on 100 mg spironolactone oral suspension. Details of study design and results are described in the individual study review 084-15.

After administration of 100 mg spironolactone suspension with a high-fat and high-calorie meal, C_{max} decreased by 22% and AUC increased by 90%. T_{max} was delayed from 0.9 ± 0.3 hours in the fasted state to 2.4 ± 1.0 hours in the fed state.

This result indicates that food has a significant impact on the bioavailability of spironolactone. The current label includes information in section 12.3 about an increase in exposure of about 100% when administered with food. Therefore, the results in this submission were expected and the current labelling language for spironolactone tablets of food effect is also applicable to spironolactone oral suspension.

2.4 Analytical section

2.4.1 Are the bioanalytical methods properly validated to measure spironolactone in plasma samples?

The plasma samples of spironolactone and its active metabolites were analyzed by (b) (4) for study 063-15; while the plasma samples of spironolactone and canrenone were analyzed by (b) (4) for study 064-15 and 084-15. Thus, two method validation reports from each analytical site and three analytical reports were submitted the division. The bioanalytical assay validation for the measurement of spironolactone in plasma samples were summarized in Table 7. All validations for the LC/MS/MS bioanalytical assay of spironolactone appear acceptable with reasonable precision and accuracy.

The analytical methods for spironolactone metabolites, canrenone and TMS, are also acceptable. However, since the metabolites were not used for the relative bioavailability evaluation between spironolactone oral suspension and the LD, the details of method validation and performance for the metabolites are only described in the individual study reviews.

Table 7. Validation of the bioanalytical assay to measure spironolactone in plasma samples.

Study	063-15	064-15 and 084-15
Method	LC-MS/MS	LC-MS/MS
Matrix	Human Plasma	Human Plasma
Extraction	Solid Phase Extraction	Liquid-liquid Extraction
LLOQ (ng/mL)	1.0 ng/mL	0.500 ng/mL
Range	1.0 to 150.0 ng/mL	0.5, 1.0, 4.0, 15.0, 60.0, 250.0, 400.0 and 500 ng/mL
QCs	1.0, 3.0, 15.0, 75.0, 125.0 and 150 ng/mL	0.500, 1.500, 230.0 and 400.0 ng/mL
Accuracy/Bias	Within-run: 92.5 to 107.1% Between run: 94.3 to 102.0%	Intra-day: 99.1% to 113.6% Inter-day: 104.1% to 108.7%
Precision (CV%)	Within-run: 0.62 to 2.94% Between run: 1.51 to 6.53%	Intra-day: 2.3% to 11.3% Inter-day: 4.4% to 7.6%
Recovery	Spironolactone: 68.6 to 69.9% Internal Standard: 70.7%	63.8% - 68.6% IS: 65.0% to 70.2%
Carry-over	No carry-over effect	≤ 1.07%
Dilution Integrity	Dilution factor of 1/5	1/5 dilution

Labeling Recommendations (Draft)

Sections 1 (Indications) and 2 (Dosage and Administration):

At this time, the applicant does not have sufficient data to support use of spironolactone oral suspension at doses above 100 mg. (b) (4)

Section 2 (Dosage and Administration): A 25% dose reduction is recommended for spironolactone oral suspension relative to the LD at each dose level between 25 mg and 100 mg. In addition, the dose of 18.75 mg (3.75 mL) is further round up to 20 mg (4mL) for the consideration of applicability during clinical practice. The details of dosing adjustment strategy for section [Dosage and Administration 2] are summarized in the table below.

<i>Aldactone® Dose (mg)</i>	<i>Adjusted Dose for Spironolactone Oral Suspension (mg)</i>	<i>Spironolactone Oral Suspension Volume (mL)</i>
25	20	4
50	37.5	7.5
100	75	15

Section 7.1 (Drugs and Supplements Increasing Serum Potassium):

This text is too ambiguous and the issue of use in patients (b) (4)

Section 7 (Drug Interactions):

The literature reference reviewed below showed that salicylates can reduce the efficacy of spironolactone; therefore, the Division proposes adding a section describing the interaction of salicylates with spironolactone as below:

“Salicylates may reduce the efficacy of spironolactone. Therefore, when CAROSPIR and salicylates are used concomitantly, CAROSPIR may need to be titrated to higher maintenance dose and the patient should be observed closely to determine if the desired effect is obtained [see Clinical Pharmacology (12.3)].”

Reviewer's Comments: 600 mg of acetylsalicylic acid given orally reduces the natriuretic effects of spironolactone in a single-dose experiment in normal subjects (Tweeddale and Ogilvie, 1973) by 13-24%, depending on the dose of spironolactone, and by 30% during chronic treatment with 25 mg spironolactone q.i.d. This observation has been confirmed by Ramsay et al. (1976). Acetylsalicylic acid probably blocks the carrier-mediated secretion of the principal spironolactone metabolite canrenone in the proximal convoluted tubule, thus blunting its diuretic effect. (Offerhaus, Pharmac. Ther. Vol. 15. pp. 69 to 78, 1981)

Section 8.7 (Use in Hepatic Impairment):

Clearance of spironolactone and its metabolites is reduced in patients with cirrhosis, leading to increases in steady-state terminal half-lives (Sungaila, I.; et al. Gastroenterology 1992; 102:1680–1685). In patients with cirrhosis, start with lowest initial dose and titrate slowly. Therefore, the Division proposes adding the following language in section *[Use in Hepatic Impairment 8.7]*:

Section 12.3 (Pharmacokinetics):

- The Division proposes adding the following language to section 12.3 to address the difference in pharmacokinetics between spironolactone oral suspension and the LD.

“CAROSPIR shows 15 to 37% higher exposure compared to spironolactone tablets, thus requiring adjustment of doses by 25%.”

- Per Guidance for Industry: Clinical Pharmacology Section Labeling Guidance from 2016, (b) (4)
[REDACTED]
- The division proposes adding ‘patients with hepatic impairment’ as a specific population in section 12.3 as following: the terminal half-life of spironolactone has been reported to be increased in patients with cirrhotic ascites. See labeling changes for section 8.7 above for rationale.
- The division proposes adding relevant drug interactions, including drugs and supplements increasing serum potassium, lithium, nonsteroidal anti-inflammatory drugs, digoxin and salicylates. See labeling changes for section 7 above for rationale.

3 Appendices

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3.1 Pilot Bioequivalence Study [Spironolactone]

Report # 063-15		Study Period 03/04/2016-03/14/2016	EDR Link \\CDSESUB1\evsprod \NDA209478\0000
Title		An open label, randomized, two treatment, two period, two sequence, crossover, single dose, oral pharmacokinetic and comparative bioavailability study of Spironolactone suspension 25 mg / 5 mL of CMP Pharma, USA with Aldactone® Spironolactone tablets 25 mg of G.D. Searle LLC, Division of Pfizer Inc. USA, in healthy, human subjects under fasting condition.	
Study Design:			
Open label, randomized, 2-treatment, 2-period, 2-sequence, crossover, single dose.			
Study Participants: 14 adult male subjects aged between 18-45 years, body mass index (BMI) within 18.5 to 24.9 kg/m².			
Group 1 Treatment Washout 7 days Reference Group 2 Reference Washout 7 days Treatment Period 1 Period 2			
Screening: Day -1		Washout: 7 days	
Period 1/2		Subjects will be housed for at least 11.00 hours prior to drug administration on Day 1 and 8, until 60.00 hours post dose in each period with at least 7 days washout between each consecutive treatment periods.	
Treatments:			

	Test product T*	Reference product R*
Dosage form	Suspension	Tablets
Drug Name	Spironolactone suspension 25 mg / 5 mL	Aldactone® spironolactone tablets 25 mg
Manufacturer	CMP Pharma, USA	G.D Searle Division of Pfizer Inc. USA
Lot Number	615241B	C111814
Manufacturing Date	24 Jun 2015	NA
Expiry Date	Jun 2016	06-16

* As per study drug receipt checklist and acknowledgement form

Source: Applicant's Clinical Study Report (063-15) (NDA 205008, Sequence 0000 (1) Section 5.3.1.2).

Sampling Times

0 (predose), 0.25, 0.50, 0.75, 1.00, 1.25, 1.50, 2.00, 2.50, 3.00, 3.50, 4.00, 5.00, 6.00, 8.00, 12.00, 24.00, 36.00, 48.00 and 60.00 hours

Analytical Methods

Method Validation Report

Analytical Site: (b) (4)

Analyte	Spironolactone	Canrenone	7α-Thiomethyl Spironolactone (TMS)
Method	LC-MS/MS	LC-MS/MS	LC-MS/MS
Matrix	Human Plasma	Human Plasma	Solid Phase Extraction
Extraction	Protein Precipitation	Protein Precipitation	Protein Precipitation
LLOQ (ng/mL)	1.0	1.5	2.5
Range (ng/mL)	1.0 to 150.0	1.5 to 450.0	2.5 to 750.0
QCs	1.0, 3.0, 15.0, 75.0, 125.0 and 150	1.5, 4.5, 45.0, 225.0, 360.0 and 450.0	2.5, 7.5, 75.0, 375.0, 625.0 and 750.0
Accuracy /Bias	Within-run: 92.5 to 107.1% Between run: 94.3 to 102.0%	Within-run: 90.1 to 102.2% Between run: 89.4 to 102.4%	Within-run: 93.8 to 105.3% Between run: 95.5 to 105.1%
Precision (CV%)	Within-run: 0.6 to 2.9% Between run: 1.5 to 6.5%	Within-run: 0.4 to 4.0% Between run: 1.7 to 5.7%	Within-run: 0.6 to 2.2% Between run: 1.2 to 4.1%
Recovery	Spironolactone: 68.6 to 69.9% Internal Standard: 70.7%	67.2 to 68.9%	64.9 to 67.3%

Carry-over	No carry-over effect	No carry-over effect	No carry-over effect
Dilution Integrity	Dilution factor of 1/5	Dilution factor of 1/5	Dilution factor of 1/5

Source: Applicant's Method Validation Report (VR-14-021-SPL) (NDA 205008, Sequence 0000 (1) Section 5.3.1.2).

Bioanalytical Report

Analyte	Spironolactone	Canrenone	7 α -Thiomethyl Spironolactone (TMS)
Method	LC-MS/MS	LC-MS/MS	LC-MS/MS
Matrix	Human Plasma	Human Plasma	Human Plasma
Extraction	Protein Precipitation	Protein Precipitation	Protein Precipitation
Range (ng/mL)	0.498 – 149.485	0.743 – 445.780	1.261-756.317
QCs (ng/mL)	1.499, 11.989, 74.928, 124.881, 224.785	2.233, 31.763, 223.332, 357.331, 669.995	3.757, 50.098, 375.735, 626.224, 1127.204
Accuracy /Bias	96.73% to 102.43%	97.27% to 101.54%	101.05% to 104.65%
Precision (CV%)	2.12% to 4.29%	2.03% to 3.54%	1.77% to 2.09%

Source: Applicant's Bioanalytical Study Report (063-15) (NDA 205008, Sequence 0000 (1) Section 5.3.1.2).

Reviewer's comments:

Precision and accuracy for QCs and calibration curves for the LC-MS/MS bioanalytical assay for spironolactone, canrenone and TMS appear acceptable. Results for incurred sample reanalysis were not submitted by the applicant.

Statistical Method:

The log-transformed pharmacokinetic parameters C_{max} , AUC_{0-t} and $AUC_{0-\infty}$ were analyzed using ANOVA model with the main effect of formulation, period and sequence as fixed effect and subject nested within sequence as random effect. The 90% confidence interval for the ratio of the geometric means of the Test drug and the Reference drug were calculated for Log transformed C_{max} , AUC_{0-t} and $AUC_{0-\infty}$ with respect to spironolactone.

Study Population :

Randomized/Completed/ Discontinued Due to AE	14 / 14 / 0
Pharmacokinetic Evaluable Population*	13

Age [Median (range)]	36.07 ± 4.25
Male/Female	14 / 0
Race (Caucasian/Black/Asian/Hispanic)	0 / 0 / 14 / 0

*Subject (b) (6) was excluded from pharmacokinetic data analysis for the LD because duplicated plasma concentrations were collected at 0.5 hour due to 15 minutes deviation of the sample at 0.25 hour at period II.

Source: Applicant's Clinical Study Report (063-15) (NDA 205008, Sequence 0000 (1) Section 5.3.1.2).

Results

Table 1. Summary Statistics for PK Parameters of spironolactone, canrenone and TMS by Treatment.

Parameters	Study 063-15					
	Spironolactone Suspension (25 mg)			Aldactone® (25 mg)		
	Spironolactone	Canrenone	TMS*	Spironolactone	Canrenone	TMS*
N	14	14	14	13	13	13
C _{max} (ng/mL)	47.4 (40.7%)	22.5 (23.0%)	165.6 (24.8%)	40.4 (35.6%)	20.0 (18.0%)	143.3 (21.0%)
AUC _{0-t} (ng*hr/mL)	76.7 (35.1%)	474.9 (15.9%)	1458.0 (34.8%)	68.8 (33.6%)	448.5 (16.5%)	1405.3 (35.8%)
AUC _{0-inf} (ng*hr/mL)	81.6 (35.0%)	742.4 (19.0%)	1763.3 (43.0%)	72.6 (33.4%)	730.5 (31.7%)	1663.9 (42.4%)
T _{max} (hr)	0.8 (34.5%)	2.5 ± 1.0	1.23 ± 0.5	0.9 (41.3%)	2.9 ± 1.0	1.4 ± 0.4
T _{1/2} (hr)	3.7 (53.0%)	45.2 ± 14.2	29.1 ± 10.8	3.4 (41.8%)	56.8 ± 57.3	27.2 ± 10
AUC _{%Extrap}	5.6 (37.6%)	33.4 (36.4%)	16.8 (9.4%)	5.0 (21.1%)	33.7 (40.7%)	12.9 (69.6%)

Source: Applicant's Clinical Study Report (063-15) (NDA 205008, Sequence 0000 (1) Section 5.3.1.2).

* Source: Reviewer's analysis.

Table 2. Summary of Statistical Analysis of Ln-transformed Pharmacokinetic Parameters of Spironolactone

PK Parameters (Unit)	Geometric Least Square Means and It's Ratio			ISCV (%)	90% CI	Power (%)
	Test Product (T)	Reference Product (R)	(T/R) (%)			
C _{max} (ng/mL)	47.360	38.901	121.744%	26.28%	101.43% - 146.13%	65.12%
AUC _{0-t} (ng.hr/mL)	76.695	67.289	113.978%	14.39%	103.02% -126.11%	97.35%
AUC _{0-inf} (ng.hr/mL)	81.556	70.786	115.214%	14.31%	104.19% - 127.41%	97.44%

T: Test product: Spironolactone suspension 25 mg / 5 mL

R: Reference product: Aldactone® spironolactone tablets 25 mg

Source: Applicant's Clinical Study Report (063-15) (NDA 205008, Sequence 0000 (1) Section 5.3.1.2).

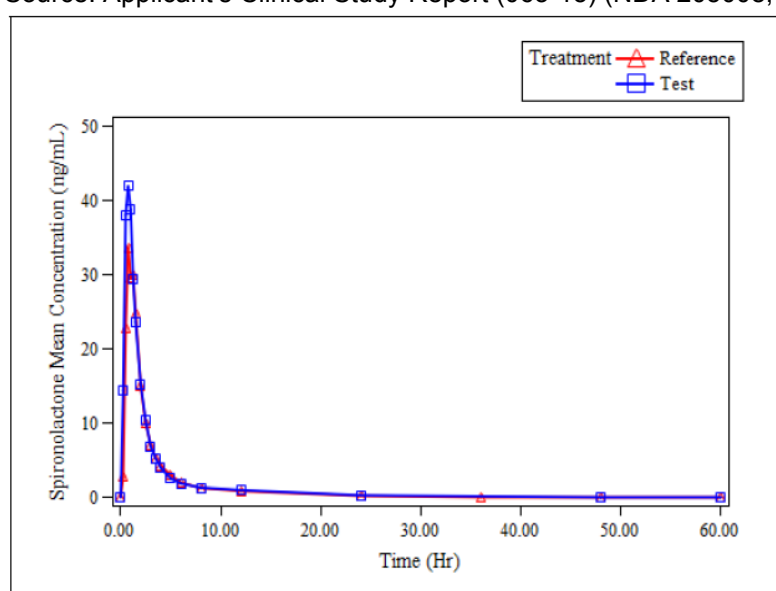


Figure 1. Mean Plasma concentration vs time profiles - treatment wise of spironolactone (Linear graph).

Source: Applicant's Clinical Study Report (063-15) (NDA 205008, Sequence 0000 (1) Section 5.3.1.2).

Reviewer's comments:

The expected T_{max} is about 1, 2.5 and 3 hours for spironolactone, canrenone and TMS, respectively. The elimination half-life of spironolactone is about 1.3-2 hours, whereas those for canrenone and TMS are about 10-45 hours. The sampling time points cover more than 5 half-lives for spironolactone, which is reasonable. However, the sampling points covers less than 2 half-lives for canrenone and TMS, which will result in an underestimated $AUC_{0-\infty}$.

Safety

Was there any death or serious adverse events? ☐ Yes ☒ No ☐ NA
Conclusion The %T/R ratio of spironolactone was 121.7%, 114.0% and 115.2% for C_{max}, AUC_{0-t} and AUC_{0-inf}, respectively. The lower and the upper limits at 90% confidence interval were 101.4% - 146.1%, 103.0% - 126.1% and 104.2% - 127.4% for C_{max}, AUC_{0-t} and AUC_{0-inf}, respectively. Spironolactone suspension 25 mg (dosed as 5 mL of 25 mg/5 mL suspension) is not bioequivalent to spironolactone tablets 25 mg. The relative bioavailability of spironolactone is approximately 15% higher for spironolactone oral suspension compared to Aldactone® tablets at 25 mg.

3.2 Pivotal Bioequivalence Study [Spironolactone]

Report # 064-15	Study Period 06/09/2016 – 06/19-2016	EDR Link \\CDSESUB1\evsprod \NDA209478\0000
Title	An open label, randomized, two treatment, two period, two sequence, crossover, single dose, oral pharmacokinetic and comparative bioavailability study of Spironolactone suspension 100 mg (20 mL of 25 mg/ 5 mL) of CMP Pharma, USA with Aldactone® (spironolactone) tablets USP, 100 mg of G.D. Searle LLC, Division of Pfizer Inc. USA, in healthy, human adult subjects under fasting condition.	
Study Design: Open label, randomized, 2-treatment, 2-period, 2-sequence, crossover, single dose. Study Participants: 56 adult male subjects aged between 18-45 years, body mass index (BMI) within 18.5 to 24.9 kg/m². Group 1 Treatment → Washout 7 days → Reference Group 2 Reference → Washout 7 days → Treatment Period 1 Period 2		
Screening: Day -1		Washout: 7 days
Period 1/2	Subjects will be housed for at least 11.00 hours prior to drug administration on Day 1 and 8, until 48.00 hours post dose in each period with at least 7 days washout between each consecutive treatment periods.	

Treatments:

	Test product T*	Reference product R*
Dosage form	Suspension	Tablets
Drug Name	Spironolactone suspension 100 mg (20 mL of 25 mg / 5 mL)	Aldactone® spironolactone tablets 100 mg
Manufacturer	CMP Pharma, USA	G.D Searle Division of Pfizer Inc. USA
Lot Number	615241B	C150002
Manufacturing Date	24 Jun 2015	NA
Expiry Date	Jun 2016	Apr 2019

* As per study drug receipt checklist and acknowledgement form

Source: Applicant's Clinical Study Report (064-15) (NDA 205008, Sequence 0000 (1) Section 5.3.1.2).

Sampling Times (PK, Plasma)

0 (predose), 0.25, 0.50, 0.75, 1.00, 1.25, 1.50, 2.00, 2.50, 3.00, 3.50, 4.00, 5.00, 6.00, 8.00, 12.00, 24.00, 36.00, 48.00 and 60.00 hours in each study period.

Analytical Methods

Analytical Site: (b) (4)

Method Validation Report

Analyte	Spironolactone	Canrenone
Method	LC-MS/MS	LC-MS/MS
Matrix	Human Plasma	Human Plasma
Extraction	Liquid-liquid extraction	Liquid-liquid extraction
LLOQ (ng/mL)	0.500 ng/mL	1.000 ng/mL
Range (ng/mL)	0.5, 1.0, 4.0, 15.0, 60.0, 250.0, 400.0 and 500	1.0, 2.0, 6.0, 20.0, 60.0, 150.0, 240.0 and 300.0
QCs (ng/mL)	0.500, 1.500, 230.0 and 400.0	1.0, 3.0, 130.0 and 230.0
Accuracy/Bias	Intra-day: 99.1% to 113.6% Inter-day: 104.1% to 108.7%	Intra-day: 87.1% to 112.4% Inter-day: 96.9% to 100.9%
Precision (CV%)	Intra-day: 2.3% to 11.3% Inter-day: 4.4% to 7.6%	Intra-day: 2.1% to 11.7% Inter-day: 2.9% to 8.1%
Recovery	63.8% - 68.6% IS: 65.0% to 70.2%	65.1% to 66.4%
Carry-over	≤ 1.07%	No carry-over effect
Dilution Integrity	1/5 dilution	1/5 dilution

Source: Applicant's Method Validation Study Report (MVR/16/004-00) (NDA 205008, Sequence 0000 (1) Section 5.3.1.2).

Bioanalytical Report

Analyte	Spironolactone	Canrenone
Method	LC-MS/MS	LC-MS/MS
Matrix	Human Plasma	Human Plasma
Extraction	Liquid-liquid extraction	Liquid-liquid extraction
Range	0.494 - 494.716	0.997 – 294.959
QCs	1.476, 230.597, 384.328	2.991, 138.460, 230.767
Accuracy/Bias	95.1- 103.2	99.6 - 100.4
Precision (CV%)	8.9 -- 12.2	5.6 - 11.0
ISR*	117 out of 224 (79.0%) samples passed.	192 out of 224 (85.7%) samples passed.

Source: Applicant's Bioanalytical Study Report (064-15) (NDA 205008, Sequence 0000 (1) Section 5.3.1.2).

Reviewer's comments:

All method validations for the LC-MS/MS bioanalytical assay for spironolactone and canrenone appear acceptable with reasonable precision and accuracy.

Statistical Method:

The log-transformed pharmacokinetic parameters C_{max} , AUC_{0-t} and $AUC_{0-\infty}$ were analyzed using ANOVA model with the main effect of formulation, period and sequence as fixed effect and subject nested within sequence as random effect. The 90% confidence interval for the ratio of the geometric means of the Test drug and the Reference drug were calculated for Log transformed C_{max} , AUC_{0-t} and $AUC_{0-\infty}$ with respect to spironolactone.

Study Population :

Randomized/Completed/ Discontinued Due to AE	56 / 56 / 0
Pharmacokinetic Evaluable Population*	56
Age [Median (range)]	29.2 ± 5.5
Male/Female	56 / 0
Race (Caucasian/Black/Asian/Hispanic)	0 / 0 / 56 / 0

Source: Applicant's Study Report (064-15) (NDA 205008, Sequence 0000 (1) Section 5.3.1.2).

Results

Table 1. Summary of Pharmacokinetic Parameters for Spironolactone

Parameters	Study 064-15			
	Spironolactone Suspension (100 mg)		Aldactone® (100 mg)	
	Spironolactone	Canrenone	Spironolactone	Canrenone
N	56	56	56	56

C _{max} (ng/mL)	127.0 (41.6%)	94.0 (28.5%)	76.7 (53.6%)	69.1 (31.8%)
AUC _{0-t} (ng*hr/mL)	290.1 (47.6%)	1901.3 (26.5%)	211.2 (56.9%)	1603.5 (27.2%)
AUC _{0-inf} (ng*hr/mL)	299.5 (50.3%)	2747.1 (31.8%)	218.6 (57.6%)	2345.5 (33.6%)
T _{max} (hr)	6.4 ± 0.3	2.6 ± 0.9	1.4 ± 0.8	3.6 ± 1.7
T _{1/2} (hr)	6.4 ± 3.3	37.9 ± 15.2	5.3 ± 2.5	38.2 ± 16.7
AUC _{%Extrap}	2.4 (61.3%)	26.1 (57.6%)	2.9 (49.8%)	26.6 (60.2%)

Source: Applicant's Study Report (064-15) (NDA 205008, Sequence 0000 (1) Section 5.3.1.2).

T: Test product: Spironolactone suspension 100 mg (20 mL of 25 mg / 5 mL)

R: Reference product: Aldactone® spironolactone tablets 100 mg

Table 2. Summary of Statistical Analysis of Ln-transformed Pharmacokinetic Parameters of Spironolactone

PK Parameters (Unit)	Geometric Least Square Means and It's Ratio			ISCV (%)	90% CI	Power (%)
	Test Product (T)	Reference Product (R)	(T/R) (%)			
C _{max} (ng/mL)	127.021	76.710	165.587%	31.08%	150.42% - 182.28%	98.45%
AUC _{0-t} (ng.hr/mL)	290.114	211.220	137.352%	33.67%	123.83% - 152.35%	97.06%
AUC _{0-inf} (ng.hr/mL)	299.509	218.630	136.994%	36.63%	122.45% - 153.26%	94.81%

Source: Applicant's Study Report (064-15) (NDA 205008, Sequence 0000 (1) Section 5.3.1.2).

T: Test product: Spironolactone suspension 100 mg (20 mL of 25 mg / 5 mL)

R: Reference product: Aldactone® spironolactone tablets 100 mg

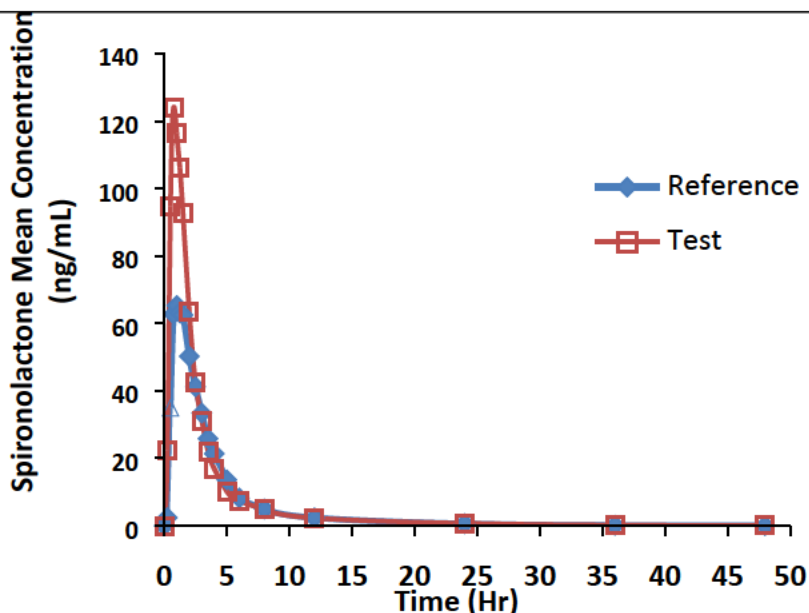


Figure 1. Mean Plasma concentration vs time profiles - treatment wise of spironolactone (Linear graph).

Source: Reviewer's plotting

Reviewer's comments:

The sampling time points cover more than 5 half-lives for spironolactone, which is reasonable. However, the sampling points covers less than 2 half-lives for canrenone, which will result in an underestimated $AUC_{0-\infty}$.

Safety

Was there any death or serious adverse events? ☐ Yes ☒ No ☐ NA

Conclusion

The %T/R ratio of spironolactone was 165.6%, 137.4% and 137.0% for C_{max} , AUC_{0-t} and $AUC_{0-\infty}$, respectively. The lower and the upper limits at 90% confidence interval were 150.4% - 182.3%, 123.8% - 152.4% and 122.5% - 153.3% for C_{max} , AUC_{0-t} and $AUC_{0-\infty}$ respectively, which were not within the standard bioequivalence limit of 80.0-125.0%.

Spironolactone suspension 100 mg (dosed as 20 mL of 25 mg/5 mL suspension) and spironolactone 100 mg tablets were not bioequivalent. The relative bioavailability of spironolactone is approximately 37% higher for spironolactone oral suspension compared to Aldactone® tablets at 100 mg.

3.3 Food Effect Study [Spironolactone]

Report # 084-15		Study Period 05/31/2016-07/10/2016	EDR Link \\CDSESUB1\evsprod\NDA209478\0000																																																												
Title	An open label, balanced, randomized, single dose, two treatment (fed Vs. fasting), two-period, two-way cross over, oral food effect study of Spironolactone suspension 100 mg (20 mL of 25 mg/ 5 mL) of CMP Pharma, USA, in healthy, human adult subjects..																																																														
Study Design: Open label, randomized, 2-treatment, 2-period, 2-sequence, crossover, single dose. Each randomized participants received 100 mg spironolactone oral suspension after one of the following treatments in 2 different treatment periods: • 10 hours overnight fasting. • Each participant finished a high-fat and high-calorie breakfast within 30 minutes prior to administration of investigational product. Study Participants: 24 adult male subjects aged between 18-45 years, body mass index (BMI) within 18.5 to 24.9 kg/m ² .																																																															
Group 1 Treatment A Washout 7 days Treatment B Group 2 Treatment B Washout 7 days Treatment A Period 1 Period 2																																																															
The composition of the food taken for fed treatment group was summarized in Table 1.																																																															
Table 1. Composition of the meal used in food-effect study 084-15.																																																															
<table><tr><th>Ingredients</th><th>Amount (g)</th><th>Energy (kcal)</th><th>Protein (kcal)</th><th>Fat (kcal)</th><th>Carbohydrates (kcal)</th></tr><tr><td>Bread</td><td>40</td><td>126.5</td><td>15.2</td><td>8.1</td><td>103.2</td></tr><tr><td>Butter</td><td>20</td><td>144</td><td>0</td><td>144</td><td>0</td></tr><tr><td>Omelet</td><td>50</td><td>129.5</td><td>24</td><td>10 3.5</td><td>2</td></tr><tr><td>Chicken Tikka</td><td>75</td><td>158.2</td><td>64</td><td>88.2</td><td>6</td></tr><tr><td>French fries</td><td>25</td><td>139</td><td>7</td><td>72</td><td>60</td></tr><tr><td>Milk (whole)</td><td>240 mL</td><td>229.68</td><td>41.28</td><td>140.4</td><td>48</td></tr><tr><td>Sugar</td><td>10</td><td>40</td><td>0</td><td>0</td><td>40</td></tr><tr><td>TOTAL</td><td></td><td>966.88</td><td>151.48</td><td>556.2</td><td>259.2</td></tr><tr><td>PERCENTAGE</td><td></td><td>100</td><td>15.67</td><td>57.52</td><td>26.8</td></tr></table>				Ingredients	Amount (g)	Energy (kcal)	Protein (kcal)	Fat (kcal)	Carbohydrates (kcal)	Bread	40	126.5	15.2	8.1	103.2	Butter	20	144	0	144	0	Omelet	50	129.5	24	10 3.5	2	Chicken Tikka	75	158.2	64	88.2	6	French fries	25	139	7	72	60	Milk (whole)	240 mL	229.68	41.28	140.4	48	Sugar	10	40	0	0	40	TOTAL		966.88	151.48	556.2	259.2	PERCENTAGE		100	15.67	57.52	26.8
Ingredients	Amount (g)	Energy (kcal)	Protein (kcal)	Fat (kcal)	Carbohydrates (kcal)																																																										
Bread	40	126.5	15.2	8.1	103.2																																																										
Butter	20	144	0	144	0																																																										
Omelet	50	129.5	24	10 3.5	2																																																										
Chicken Tikka	75	158.2	64	88.2	6																																																										
French fries	25	139	7	72	60																																																										
Milk (whole)	240 mL	229.68	41.28	140.4	48																																																										
Sugar	10	40	0	0	40																																																										
TOTAL		966.88	151.48	556.2	259.2																																																										
PERCENTAGE		100	15.67	57.52	26.8																																																										
Source: Applicant's Response to Information Request (Mar 21, 2017) (NDA 205008, Sequence 0007 (8) Section 1.2).																																																															

Screening: Day -1		Washout: 7 days
Period 1/2	Subjects will be housed for at least 11.00 hours prior to drug administration on Day 1 and 8, until 48.00 hours post dose in each period with at least 7 days washout between each consecutive treatment periods.	
Treatments:		
	Treatment A	Treatment B
Dosage form	Suspension	Suspension
Drug Name	Spironolactone suspension 100 mg (20 mL of 25 mg / 5 mL) under fed condition	Spironolactone suspension 100 mg (20 mL of 25 mg / 5 mL) under fasting condition
Manufacturer	CMP Pharma, USA	CMP Pharma, USA
Lot Number	615241B	615241B
Manufacturing Date	24 Jun 2015	24 Jun 2015
Expiry Date	Jun 2016	Jun 2016
Source: Applicant's Study Report (084-15) (NDA 205008, Sequence 0000 (1) Section 5.3.1.2).		
Sampling Times		
0 (predose), 0.25, 0.50, 0.75, 1.00, 1.25, 1.50, 2.00, 2.50, 3.00, 3.50, 4.00, 5.00, 6.00, 8.00, 12.00, 24.00, 36.00, 48.00 and 60.00 hours		
Method Validation		
Analytical Site: (b) (4)		
Method Validation Report		
Same as Pivotal Study (064-15)		
Bioanalytical Report		
Analyte	Spironolactone	Canrenone
Method	LC-MS/MS	LC-MS/MS
Matrix	Human Plasma	Human Plasma
Extraction	Liquid-liquid extraction	Liquid-liquid extraction
Range (ng/mL)	0.500, 1.001, 4.004, 15.130, 60.042, 250.173, 398.365 and 501.719	1.001, 2.001, 5.999, 19.997, 59.764, 148.666, 239.785 and 299.731
QCs (ng/mL)	1.473, 230.122, 383.537	2.954, 139.336, 232.227
Accuracy/Bias	97.9% -- 103.0%	97.9% -- 102.4%
Precision (CV)	8.8% -- 13.8%	3.5% -- 8.0%
ISR	77 out of 100 (77.0%) samples passed.	84 out of 100 (84%) samples passed.

Source: Applicant's Bioanalytical Report (084-15) (NDA 205008, Sequence 0000 (1) Section 5.3.1.2).

Reviewer's Comments:

All method validations for the LC-MS/MS bioanalytical assay for spironolactone and canrenone appear acceptable with reasonable precision and accuracy.

Statistical Method:

The log-transformed pharmacokinetic parameters C_{max}, AUC_{0-t} and AUC_{0-∞} were analyzed using ANOVA model with the main effect of formulation, period and sequence as fixed effect and subject nested within sequence as random effect. The 90% confidence interval for the ratio of the geometric means of the Test drug and the Reference drug were calculated for Log transformed C_{max}, AUC_{0-t} and AUC_{0-∞} with respect to spironolactone.

Study Population :

Randomized/Completed/ Discontinued Due to AE	24 / 23 / 0
Pharmacokinetic Evaluable Population*	23
Age [Median (range)]	29.5 ± 7.1 years
Male/Female	24 / 0
Race (Caucasian/Black/Asian/Hispanic)	0 / 0 / 24 / 0

* Subject (b) (6) did not report to facility for check-in during period II, hence considered as dropout from the study. Thus, the pharmacokinetic analysis was performed on concentration data from all 23 subjects completed the study.

Source: Applicant's Study Report (084-15) (NDA 205008, Sequence 0000 (1) Section 5.3.1.2).

Results

Table 1. Summary of pharmacokinetic parameters for spironolactone and canrenone.

Parameters	Study 084-15			
	Treatment A		Treatment B	
	Spironolcatone	Canrenone	Spironolcatone	Canrenone
N	23	23	23	23
C _{max} (ng/mL)	92.3 (42.1%)	73.3 (26.2%)	117.4 (43.8%)	96.2 (31.7%)
AUC _{0-t} (ng*hr/mL)	468.5 (35.4%)	1925.3 (19.9%)	242.9 (42.0%)	1822.5 (21.2%)
AUC _{0-inf} (ng*hr/mL)	477.9 (34.9%)	2787.6 (19.0%)	250.8 (41.4%)	2454.6 (21.3%)
T _{max} (hr)	2.4 ± 1.0	4.9 ± 2.0	0.9 ± 0.3	2.4 ± 0.7

$T_{1/2}$ (hr)	7.7 ± 3.5	36.8 ± 13.3	6.1 ± 2.7	33.0 ± 12.0
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Source: Reviewer's analysis.

Treatment A: Spironolactone suspension 100 mg (20 mL of 25 mg/ 5 mL) of CMP Pharma, USA [under fed conditions]

Treatment B: Spironolactone suspension 100 mg (20 mL of 25 mg/ 5 mL) of CMP Pharma, USA [under fasting conditions]

Table 2. Summary of Statistical Analysis of Ln-transformed Pharmacokinetic Parameters of Spironolactone

PK parameter	Geometric Least square Mean (A)	Geometric Least Square Mean (B)	Ratio (A/B)	90% CI [Lower, Upper]
C_{max} (ng/mL)	4.524	4.767	78.431%	69.57% - 88.42%
AUC_{0-t} (ng.hr/mL)	6.150	5.494	192.760%	175.74% - 211.43%
AUC_{0-inf} (ng.hr/mL)	6.170	5.525	190.483%	173.94% - 208.60%

Source: Applicant's Study Report (084-15) (NDA 205008, Sequence 0000 (1) Section 5.3.1.2).

Treatment A: Spironolactone suspension 100 mg (20 mL of 25 mg/ 5 mL) of CMP Pharma, USA [under fed conditions]

Treatment B: Spironolactone suspension 100 mg (20 mL of 25 mg/ 5 mL) of CMP Pharma, USA [under fasting conditions]

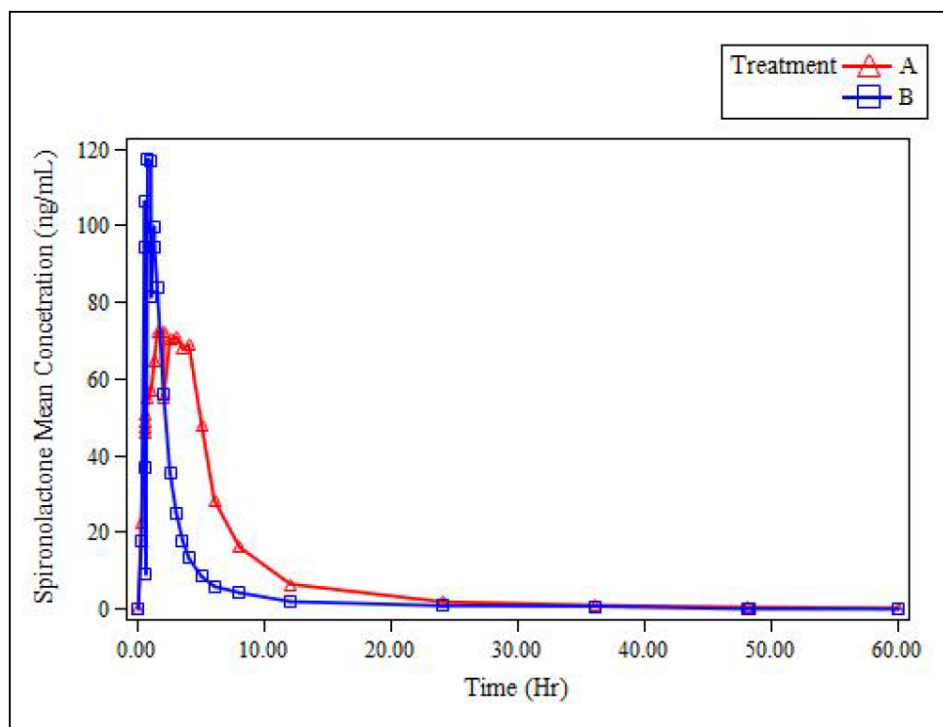


Figure 1. Mean Plasma concentration vs time profiles for spironolactone (linear graph)

Source: Applicant's Study Report (084-15) (NDA 205008, Sequence 0000 (1) Section 5.3.1.2).

Treatment A: Spironolactone suspension 100 mg (20 mL of 25 mg/ 5 mL) of CMP Pharma, USA [under fed conditions]

Treatment B: Spironolactone suspension 100 mg (20 mL of 25 mg/ 5 mL) of CMP Pharma, USA [under fasting conditions]

Reviewer's Comments:

The sampling time points for spironolactone cover more than 5 half-lives, which is reasonable. However, the sampling points covers less than 2 half-lives for canrenone, which will result in an underestimated $AUC_{0-\infty}$.

Safety

Was there any death or serious adverse events? ☐ Yes ☒ No ☐ NA

Conclusion

The test product was safe and well tolerated by all the study subjects, when administered in a single dose. Food effect of the Spironolactone Suspension 100 mg (20 mL of 25 mg/ 5 mL) of CMP Pharma, USA., is established since the 90% confidence intervals for relative mean AUC_{0-t} , $AUC_{0-\infty}$ and C_{max} of the treatment A and B were not within 80.0% to 125.0% for log transformed data of Spironolactone.

High-fat, high calories food increased the bioavailability of spironolactone oral suspension by approximately 90%.

Comments

- The applicant incorrectly labeled 'treatment A' and 'treatment B' in Table 1-4 and Table 10-11 in the study report. The reviewer recalculated the PK-parameters for spironolactone and canrenone based on submitted datasets, and the results were presented in the results section.

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/s/

XIAOLEI N PAN
06/05/2017

MARTINA D SAHRE
06/05/2017

CLINICAL PHARMACOLOGY FILING FORM

Application Information

NDA/BLA Number	209478	SDN	0000, 0001, 0002
Applicant	CMP Pharma, Inc.	Submission Date	10/04/2016
Generic Name	Spironolactone	Brand Name	CaroSpir®
Drug Class	Aldosterone Antagonist		
Indication	- Essential hypertension - Severe heart failure		
Dosage Regimen			
Dosage Form	Oral Suspension (25 mg/5 mL)	Route of Administration	PO
OCP Division	DCP 1	OND Division	DCRP
OCP Review Team	Primary Reviewer(s)	Secondary Reviewer / Team Leader	
Division	Xiaolei Pan	Martina Sahre / Sudharshan Hariharan	
Pharmacometrics	-	-	
Genomics	-	-	
Review Classification	☒ Standard ☐ Priority ☐ Expedited		
Filing Date	12/3/2016	74-Day Letter Date	12/17/2016
Review Due Date	6/5/2017	PDUFA Goal Date	8/4/2017

Application Fileability

Is the Clinical Pharmacology section of the application fileable?

☒ Yes

☐ No

Are there any potential review issues/ comments to be forwarded to the Applicant in the 74-day letter?

☒ Yes

☐ No

- The results of the pivotal PK Study 064-15 indicate that CaroSpir® oral suspension and the reference listed drug (Aldactone® tablets) are not bioequivalent. Provide a justification for bridging to the efficacy and safety of the reference listed drug that addresses these findings. Please also address the clinical

relevance of the 66% increase in C_{max} (relative to Aldactone®) and 37% increase in AUC (relative to Aldactone®) seen in Study 064-15 and the differences in relative bioavailability observed in Studies 063-15 and 064-15.

2. Provide statistical analysis plans for all studies. Currently, all report Appendices 16.1.9 are labeled to contain the Documentation of Statistical Methods, but we can only locate the output from analyses in this document. Please provide the source code and analytical parameters for the SAS and WinNonlin analyses. If this has already been submitted, please provide information on the specific location within the submission.

Please submit the requested information no later than January 13, 2017.

Is there a need for clinical trial(s) inspection?

☒ Yes

☐ No

The sponsor conducted two BE studies, a pilot study (063-15, comparing to 25 mg RLD) and a pivotal study 064-15 (comparing to 100 mg RLD). The two studies showed significant differences in relative BA. If possible, we ask that analytical and clinical sites for both studies are inspected. If that is not possible, then we ask that the pivotal study (064-15) sites are inspected.

Clinical Pharmacology Package

Tabular Listing of All Human Studies ☒ Yes ☐ No Clinical Pharmacology Summary ☐ Yes ☒ No

Bioanalytical and Analytical Methods ☒ Yes ☐ No Labeling ☒ Yes ☐ No

Clinical Pharmacology Studies

Study Type	Count	Comment(s)
In Vitro Studies		
☐ Metabolism Characterization		
☐ Transporter Characterization		
☐ Distribution		
☐ Drug-Drug Interaction		
In Vivo Studies		
Biopharmaceutics		
☐ Absolute Bioavailability		
☒ Relative Bioavailability	2	
☐ Bioequivalence		
☒ Food Effect	1	
☐ Other		
Human Pharmacokinetics		
Healthy Subjects	☐ Single Dose	
	☐ Multiple Dose	
Patients	☐ Single Dose	
	☐ Multiple Dose	
☐ Mass Balance Study		
☐ Other (e.g. dose proportionality)		

Intrinsic Factors				
☐ Race				
☐ Sex				
☐ Geriatrics				
☐ Pediatrics				
☐ Hepatic Impairment				
☐ Renal Impairment				
☐ Genetics				
Extrinsic Factors				
☐ Effects on Primary Drug				
☐ Effects of Primary Drug				
Pharmacodynamics				
☐ Healthy Subjects				
☐ Patients				
Pharmacokinetics/Pharmacodynamics				
☐ Healthy Subjects				
☐ Patients				
☐ QT				
Pharmacometrics				
☐ Population Pharmacokinetics				
☐ Exposure-Efficacy				
☐ Exposure-Safety				
Total Number of Studies:		In Vitro		In Vivo
Total Number of Studies to be Reviewed:				
				3
				3

Criteria for Refusal to File (RTF)		
RTF Parameter	Assessment	Comments
1. Did the applicant submit bioequivalence data comparing to-be-marketed product(s) and those used in the pivotal clinical trials?	☐ Yes ☐ No ☒ N/A	
2. Did the applicant provide metabolism and drug-drug interaction information? (Note: RTF only if there is complete lack of information)	☐ Yes ☐ No ☒ N/A	
3. Did the applicant submit pharmacokinetic studies to characterize the drug product, or submit a waiver request?	☒ Yes ☐ No ☐ N/A	
4. Did the applicant submit comparative bioavailability data between proposed drug product and reference product for a 505(b)(2) application?	☒ Yes ☐ No ☐ N/A	
5. Did the applicant submit data to allow the evaluation of the validity of the analytical assay for the moieties of interest?	☒ Yes ☐ No ☐ N/A	
6. Did the applicant submit study reports/rationale to support dose/dosing interval and dose adjustment?	☐ Yes ☒ No ☐ N/A	
7. Does the submission contain PK and PD analysis datasets and PK and PD parameter datasets for each primary study that supports items 1 to 6 above (in .xpt format if data are submitted electronically)?	☒ Yes ☐ No ☐ N/A	
8. Did the applicant submit the module 2 summaries (e.g. summary-clin-pharm, summary-biopharm, pharmkin-written-summary)?	☐ Yes ☒ No ☐ N/A	
9. Is the clinical pharmacology and biopharmaceutics section of the submission legible, organized, indexed and paginated in a manner to allow substantive review to begin? If provided as an electronic submission, is the electronic submission searchable, does it have appropriate hyperlinks and do the hyperlinks work leading to appropriate sections, reports, and appendices?	☒ Yes ☐ No ☐ N/A	
Complete Application 10. Did the applicant submit studies including study reports, analysis datasets, source code, input files and key analysis output, or justification for not conducting studies, as agreed to at the pre-NDA or pre-BLA meeting? If the answer is 'No', has the sponsor submitted a justification that was previously agreed to before the NDA submission?	☒ Yes ☐ No ☐ N/A	This is a 505(b)(2) application, hence no pre-NDA/BLA meeting occurred. We request the statistical analysis plan that contained source code and input files in WinNonlin and SAS to obtain the results.

Criteria for Assessing Quality of an NDA (Preliminary Assessment of Quality) Checklist		
Data		
1. Are the data sets, as requested during pre-submission discussions, submitted in the appropriate format (e.g., CDISC)?	☒ Yes ☐ No ☐ N/A	
2. If applicable, are the pharmacogenomic data sets submitted in the appropriate format?	☐ Yes ☐ No ☒ N/A	
Studies and Analysis		
3. Is the appropriate pharmacokinetic information submitted?	☒ Yes ☐ No ☐ N/A	
4. Has the applicant made an appropriate attempt to determine reasonable dose individualization strategies for this product (i.e., appropriately designed and analyzed dose-ranging or pivotal studies)?	☐ Yes ☐ No ☒ N/A	
5. Are the appropriate exposure-response (for desired and undesired effects) analyses conducted and submitted as described in the Exposure-Response guidance?	☐ Yes ☐ No ☒ N/A	
6. Is there an adequate attempt by the applicant to use exposure-response relationships in order to assess the need for dose adjustments for intrinsic/extrinsic factors that might affect the pharmacokinetic or pharmacodynamics?	☐ Yes ☐ No ☒ N/A	
7. Are the pediatric exclusivity studies adequately designed to demonstrate effectiveness, if the drug is indeed effective?	☐ Yes ☐ No ☒ N/A	
General		
8. Are the clinical pharmacology and biopharmaceutics studies of appropriate design and breadth of investigation to meet basic requirements for approvability of this product?	☒ Yes ☐ No ☐ N/A	
9. Was the translation (of study reports or other study information) from another language needed and provided in this submission?	☐ Yes ☒ No ☐ N/A	

This is a representation of an electronic record that was signed electronically and this page is the manifestation of the electronic signature.

/s/

XIAOLEI N PAN
12/06/2016

MARTINA D SAHRE
12/06/2016

The review was originally filed in DARRTS on 12/1/2016. The primary reviewer (Pan) and I noticed that one check-box for a question on whether there were comments to be sent to the sponsor was incorrectly marked "No". This version corrects the check-box value to "Yes" and comments had been relayed to the Project Manager prior to 12/1/2016.